

Course Code: CSE1008	Course Title: Theory of Computation	TPC	4	0	4
Version No.	2.0				
Course Pre-requisites/ Co-requisites	None				
Anti-requisites (if any).	None				
Objectives:	1. To understand the essential mathematical foundations of automata theory. 2. To design, finite state automata and the equivalent regular expression for any given pattern. 3. To analyze and design Context Free Grammar, Pushdown Automata and Turing Machine. 4. To understand the difference between decidability and undesirability.				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement				POs/ PEOs/PSOs
CO1	Understand the abstract machines, computation and basic properties of formal languages, finite automata				PO1, PO3
CO2	Familiarity with regular language, regular expression, context-free grammars and can make grammars to produce strings from a specific language				PO1, PO2, PO3
CO3	Can differentiate regular, context-free and recursively enumerable languages and can compare and analyse to different computational models.				PO3, PO4, PO5
CO4	Acquire concepts relating to computational models including decidability and intractability and can identify limitations of some computational models and possible methods of proving them				PO3, PO4, PO5
Total Hours of Instructions: 60					
Module No. 1	Finite Automata (FA)				11 Hours
Mathematical preliminaries and notations-Finite Automata-Deterministic Finite Automata – Non-Deterministic Finite Automata and equivalence with DFA - Epsilon transitions – Minimization of Finite Automata and its applications.					
Module No. 2	Regular Expressions (RE)				9 Hours
Definition, Operators of regular expression and their precedence- Algebraic laws for Regular expressions and Kleene's Theorem - Regular expression to FA- DFA to Regular expression- Pumping Lemma for regular Languages					
Module No. 3	RE & Context Free Grammar (CFG)				9 Hours
Closure properties of Regular Languages- Decision properties of Regular Languages -Context-Free Grammar (CFG) – Derivation Trees – Ambiguity in Grammars and Languages					
Module No. 4	Pushdown Automata (PDA)				11 Hours
Definition, Graphical Notation, Instantaneous Descriptions of PDA- Acceptance by Final state, Acceptance by empty stack, Deterministic PDA- CFG to PDA - PDA to CFG					
Module No. 5	Normal forms of CFG				9 Hours
Normal forms for CFGs: CNF and GNF, Closure properties of CFLs, Decision Properties of CFLs: Emptiness, Finiteness and Membership, Pumping lemma for CFLs					
Module No. 6	Turing Machine				11 Hours

Turing machines (TM): Basic model, definition and representation, Instantaneous Description, Language acceptance by TM- Undecidable problems about TMs-Post correspondence problem (PCP)- Modified PCP- Introduction to recursive function theory - Introductory ideas on Time complexity of deterministic and nondeterministic Turing machines

Text Books

1. J.E.Hopcroft, R.Motwani and J.D Ullman, “Introduction to Automata Theory, Languages and Computations”, Pearson Education, 3rd Edition, 2013.

References

1. Micheal Sipser, “Introduction of the Theory and Computation”, Cengage Learning, 3rd edition, 2014.
2. Martin J. C., “Introduction to Languages and Theory of Computations”, McGraw Higher Ed, 3rd edition, 2009.
3. K.L.P. Mishra and N.Chandrasekaran, “Theory of Computer Science: Automata, Languages and Computation”, PHI, Third edition, 2009.
4. Papadimitriou, C. and Lewis, C. L., “Elements of the Theory of Computation”, Pearson, 2nd edition, 2015.

Mode of Evaluation	Continuous Assessment Tests-60%, Practical Assesment-40%	
	Continuous Assessment Test-1	20%
	Continuous Assessment Test-2	20%
	Final Assessment Test	20%
	Cumulative class Assessments	20%
	Assignment	20%
Recommended by the Board of Studies on	12th BoS 29-04-2023	
Date of Approval by the Academic Council	10 th AC 01.06.2023	